

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

Practical No : 14 and 15

Aim 14 : Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).

CODE :

```
import java.time.LocalDate
import scala.util.Random

object omwala_M2_pract14 {

  case class FootballData(date: LocalDate, goals: Int)

  def main(args: Array[String]): Unit = {

    val startDate = LocalDate.of(2026, 9, 1)
    val random = new Random()

    val data = (0 until 30).map { day =>
      val date = startDate.plusDays(day)
      val goals = random.nextInt(7)
      FootballData(date, goals)
    }

    println("Daily Football Goals - September 2026")
    println("-----")

    data.foreach { d =>
      println(s"${d.date} : ${d.goals} goals")
    }

    val totalGoals = data.map(_goals).sum
    val averageGoals = totalGoals.toDouble / data.length
    val minimumGoals = data.map(_goals).min
    val maximumGoals = data.map(_goals).max

    println()
    println("Basic Time Series Analysis")
    println("-----")
    println(s"Total Goals : $totalGoals")
    println(s"Average Goals : $averageGoals%.2f")
    println(s"Minimum Goals : $minimumGoals")
    println(s"Maximum Goals : $maximumGoals")
  }
}
```

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OUTPUT :

```
Run omwala_M2_pract14 x

"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
Daily Football Goals - September 2026
-----
2026-09-01 : 3 goals
2026-09-02 : 0 goals
2026-09-03 : 5 goals
2026-09-04 : 1 goals
2026-09-05 : 0 goals
2026-09-06 : 0 goals
2026-09-07 : 3 goals
2026-09-08 : 6 goals
2026-09-09 : 5 goals
2026-09-10 : 1 goals
2026-09-11 : 0 goals
2026-09-12 : 1 goals
2026-09-13 : 0 goals
2026-09-14 : 4 goals
2026-09-15 : 3 goals
2026-09-16 : 0 goals
2026-09-17 : 4 goals
2026-09-18 : 0 goals
2026-09-19 : 0 goals
2026-09-20 : 6 goals
2026-09-21 : 1 goals
2026-09-22 : 1 goals
2026-09-23 : 2 goals
2026-09-24 : 4 goals
2026-09-25 : 6 goals
2026-09-26 : 3 goals
2026-09-27 : 4 goals
2026-09-28 : 6 goals
2026-09-27 : 4 goals
2026-09-28 : 6 goals
2026-09-29 : 6 goals
2026-09-30 : 1 goals

Basic Time Series Analysis
-----
Total Goals : 76
Average Goals : 2.53
Minimum Goals : 0
Maximum Goals : 6

Process finished with exit code 0
```

Aim 15: Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3 (e.g., [1, 1^2, 1^3, 2, 2^2, 2^3, 3, 3^2, 3^3]).

CODE :

```
object omwala_M2_pract15 {

  def main(args: Array[String]): Unit = {

    val numbers = List(85, 91, 119)
```

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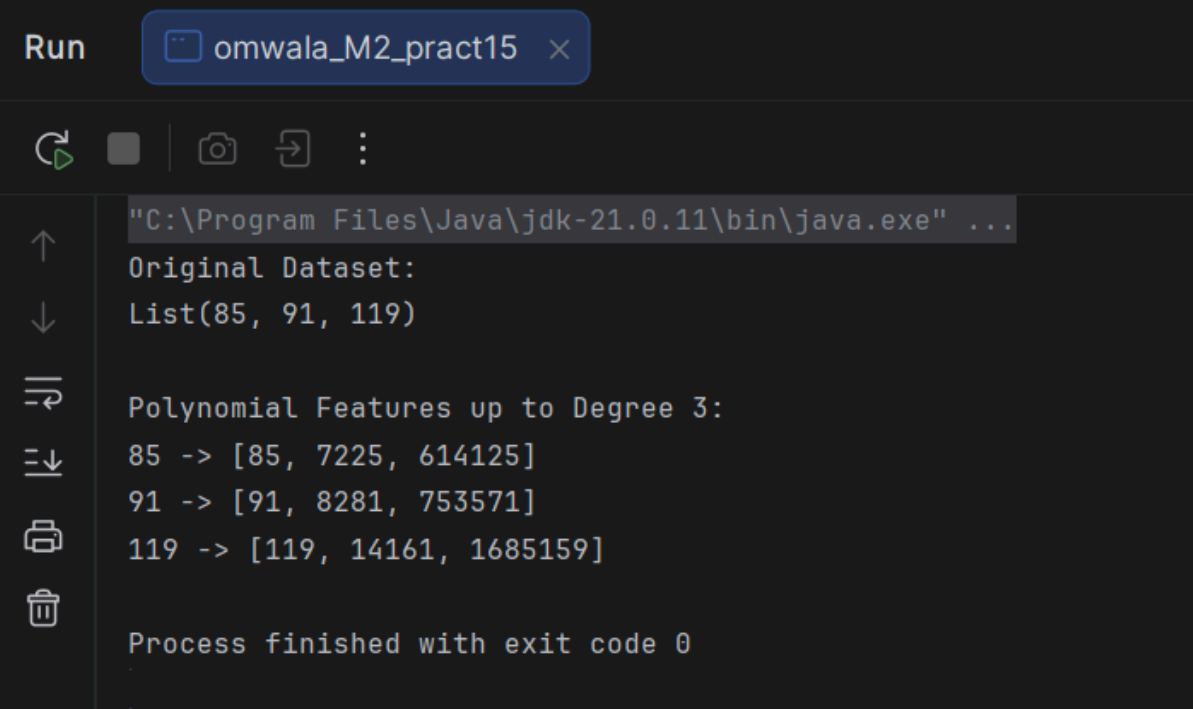
```
println("Original Dataset:")
println(numbers)

println("\nPolynomial Features up to Degree 3:")

numbers.foreach { x =>
  val x1 = x
  val x2 = x * x
  val x3 = x * x * x

  println(s"$x -> [x1, $x2, $x3]")
}
}
```

OUTPUT :



```
Run omwala_M2_pract15 x

"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...

Original Dataset:
List(85, 91, 119)

Polynomial Features up to Degree 3:
85 -> [85, 7225, 614125]
91 -> [91, 8281, 753571]
119 -> [119, 14161, 1685159]

Process finished with exit code 0
```